

AKASH SINGH

Aspiring Data Scientist | Computer Vision & Machine Learning
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PROFILE

Computer Science & Engineering diploma student who led the end-to-end build of a live, AI-powered computer-vision application during an internship — from pose-detection engine to cloud deployment. Comfortable with Python, MediaPipe, and the fundamentals of machine learning, image and video processing, and web development. Eager to apply and grow these skills on real data problems and contribute to data-driven products.

TECHNICAL SKILLS

Programming: Python, C, JavaScript

Data Science & Computer Vision: MediaPipe (pose detection), OpenCV, NumPy, image & video processing, pose-angle analysis, Machine Learning (foundations)

App & Web Development: Streamlit, HTML, CSS, JavaScript

Tools & Deployment: Git & GitHub, Streamlit Community Cloud (deployment)

EXPERIENCE

AI / Machine Learning Intern

June 2026 – July 2026

Tech Sarthi AI • Internship

- Led end-to-end development of “Posture Expert” within a 6-member team project — independently built the pose-detection engine, angle-analysis logic, app interface, and cloud deployment, and produced the project documentation and final presentation.
- Implemented pose detection with MediaPipe and computed 10 joint angles per shot, comparing each against shot-specific ideal ranges to score form and flag errors.
- Designed a scoring system producing an overall form score plus category breakdowns (Legs & Base, Arms & Racket, Torso & Balance), with prioritized, corrective coaching feedback.
- Built video support with adaptive frame sampling and automatic contact-frame detection to keep processing fast on long clips.

PROJECTS

Posture Expert — AI Tennis Shot Analyzer

Python • MediaPipe • Streamlit

Live app: ai-lawn-tennis-posture-analyzer-aakashhsinghh.streamlit.app | Code: github.com/akashhsinghh

- End-to-end computer-vision web app that analyzes a tennis player’s shot from a photo or video and returns an instant, data-driven coaching report. Sole developer within a 6-member team project.
- Detects body landmarks with MediaPipe and measures 10 joint angles (knees, arm lift, elbows, wrists, torso lean), evaluated against ideal ranges that adapt to the selected shot type and player handedness.
- Outputs a 0–100 form score with per-category sub-scores and a ranked “fix these first” list quantifying each error (e.g. knee 22° below ideal), alongside a skeleton-overlay visualization.
- Handles video via adaptive frame sampling and automatic contact-frame detection; deployed live on Streamlit Community Cloud.

EDUCATION

Diploma in Computer Science & Engineering

2025 – Present

Government Polytechnic, Gaya, India

Secondary School (Class 10)

Completed 2025

DAV Public School, Mohania, India

CERTIFICATIONS

- Python — Kaggle (2026)
- IT Essentials — Cisco Networking Academy